1. Write a C program to traverse a 2D matrix and print the values until a negative number is found. Use break statement.

main.c

```c
#include <stdio.h>

int main() {
    int rows, cols;
    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    int matrix[rows][cols];
    printf("Enter matrix elements:\n");
    for(int i = 0; i < rows; i++) {
        for(int j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }
    printf("Matrix elements until negative number:\n");
    for(int i = 0; i < rows; i++) {
        for(int j = 0; j < cols; j++) {
            if(matrix[i][j] < 0) {
                break;
            }
            printf("%d ", matrix[i][j]);
        }
        printf("\n");
        int flag = 0;
        for(int j = 0; j < cols; j++) {
            if(matrix[i][j] < 0) {
                flag = 1;
                break;
            }
        }
        if(flag == 1) {
            break;
        }
    }

    return 0;
}
```

Output

```
Enter number of rows and columns: 2 2
Enter matrix elements:
1
2

3
4
Matrix elements until negative number:
1 2
3 4


=== Code Execution Successful ===
```

2. Write a C program to print the numbers from 1 to 100 but avoid printing multiples of 3 and number containing digit 5(5,15,25,35,..). Use continue in the program.

main.c

```c
#include <stdio.h>

int main() {
    int i, temp, hasFive;

    for(i = 1; i <= 100; i++) {
        if(i % 3 == 0) {
            continue;
        }
        temp = i;
        hasFive = 0;

        while(temp > 0) {
            if(temp % 10 == 5) {
                hasFive = 1;
                break;
            }
            temp /= 10;
        }

        if(hasFive) {
            continue;
        }

        printf("%d ", i);
    }

    return 0;
}
```

Output

```
1 2 4 7 8 10 11 13 14 16 17 19 20 22 23 26 28 29 31 32 34 37 38 40 41 43 44 46 47 49 61 62 64 67 68
    70 71 73 74 76 77 79 80 82 83 86 88 89 91 92 94 97 98 100

=== Code Execution Successful ===
```

3. Write a C program that performs division of two user inputs and uses goto to handle errors like division by zero or invalid input by redirecting control for re-entry.

main.c

```c
#include <stdio.h>

int main() {
    float num1, num2, result;
    int status;

start:

    printf("Enter two numbers: ");
    status = scanf("%f %f", &num1, &num2);

    if(status != 2) {
        printf("Invalid input! Please enter numeric values.\n");
        while(getchar() != '\n');

        goto start;
    }
    if(num2 == 0) {
        printf("Error: Division by zero is not allowed.\n");
        goto start;
    }
    result = num1 / num2;
    printf("Result = %.2f\n", result);

    return 0;
}
```

Output

```
Enter two numbers: 23
03
Result = 7.67


=== Code Execution Successful ===
```

4. Write a C program for a number guessing game where the loop continues for wrong guesses, breaks on a correct guess, and exits the program using return if the user enters -1.

main.c

```c
#include <stdio.h>

int main() {
    int secret = 7;
    int guess;

    printf("Number Guessing Game\n");
    printf("Guess the number (enter -1 to exit):\n");

    while(1) {
        printf("Enter your guess: ");
        scanf("%d", &guess);
        if(guess == -1) {
            printf("You chose to exit the game.\n");
            return 0;
        }
        if(guess == secret) {
            printf("Congratulations! You guessed the correct number.\n");
            break;
        }
        if(guess < secret) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }
    }

    return 0;
}
```

Output

```
Number Guessing Game
Guess the number (enter -1 to exit):
Enter your guess: -1
You chose to exit the game.


=== Code Execution Successful ===
```

5. Write a C program that iterates through an array and, within a loop, uses continue to

skip negative elements, break when a zero is encountered, and return to exit the program if an element greater than 100 is found, then print the resulting behavior for the array {10, -5, 20, 0, 150, 30}.

main.c

Share Run

```c
#include <stdio.h>

int main() {
    int arr[] = {10, -5, 20, 0, 150, 30};
    int i;

    printf("Processing array elements:\n");

    for(i = 0; i < 6; i++) {
        if(arr[i] > 100) {
            printf("Element greater than 100 found (%d). Exiting program.\n", arr[i]);
            return 0;
        }
        if(arr[i] < 0) {
            continue;
        }
        if(arr[i] == 0) {
            printf("Zero encountered. Stopping loop.\n");
            break;
        }
        printf("%d ", arr[i]);
    }

    printf("\nProgram ended.\n");

    return 0;
}
```

Output

```
Processing array elements:
10 20 Zero encountered. Stopping loop.

Program ended.


=== Code Execution Successful ===
```